

Khrystal Aquino Cruz

Potsdam, NY 13699 - (787) 560-1888 - aquinokm@clarkson.edu

EDUCATION

Clarkson University - Bachelor of Science in Mechanical Engineering

Spring 2028

• Engineering Design • CAD Design • Statics • Materials Science • Shop Skills • Welding Procedures

Potsdam, NY

Building Skills NY / Commonpoint - Construction Training

• OSHA-30, SST-10, NCCER Core 6, Green Building Training, Silica Awareness, F-60, SCA-211, 4-Hour Flagger

ENGINEERING PROJECTS

Mechanical Component Design | SolidWorks

- Designed and modeled engine components and a jack stand using SolidWorks.
- Created detailed part models, assemblies, and technical drawings while applying engineering design principles.
- Evaluated dimensions, fit, and component functionality through iterative design improvements.

Whiteface Mountain Terrain Modeling & Additive Manufacturing | MATLAB & 3D Printing

- Developed MATLAB code to generate a three-dimensional terrain model of Whiteface Mountain using elevation and dimensional data.
- Converted computational output into a manufacturable model and produced a physical prototype using 3D printing.
- Applied programming, data visualization, and additive manufacturing techniques to create an accurate scaled representation.

Custom 3D Printed Recognition Award

- Designed and fabricated a custom Frankenstein-themed award for a university professor using additive manufacturing techniques.
- Applied design-for-manufacturing principles throughout the development process.

Arduino Automatic Traffic Light System

- Designed and programmed an Arduino-based automatic traffic light system using microcontroller logic and electronic components.
- Developed timing sequences to simulate real-world traffic signal operation.
- Integrated LEDs and circuit components while applying fundamental programming and electrical concepts.
- Tested and refined system performance through troubleshooting and iterative design improvements.

EXPERIENCE

Commonpoint

February 2025 - August 2025

Apprentice

Bronx, NY

- Coordinated project documentation, scheduling activities, and compliance records for construction and community development projects.
- Managed certification tracking and workforce documentation to support project readiness.
- Communicated with clients, contractors, and team members to support daily operations and project coordination.
- Utilized bilingual communication skills to assist English- and Spanish-speaking stakeholders.
- Worked effectively in fast-paced environments requiring adaptability, organization, and attention to detail.

Clarkson University

October 2023 - December 2024

Social Event Coordinator / Residential Assistant / Student Life Assistant

Potsdam, NY

- Coordinated logistics, scheduling, safety planning, and operational support for campus events and recreational programs.
- Collaborated with multiple university departments to organize events, manage resources, and ensure successful program execution.
- Served as a primary point of contact for students, providing guidance, mentorship, conflict resolution, and resource coordination.
- Developed and maintained event timelines, schedules, registration records, and budget tracking systems.
- Managed equipment usage and operational documentation while ensuring activities remained organized and on schedule.
- Fostered an inclusive residential community by supporting first-year students through academic, social, and personal transitions.

TECHNICAL SKILLS

CAD & Engineering Software:

SolidWorks (Parts, Assemblies, Technical Drawings), MATLAB, OpenRocket, C++, Inkscape

Manufacturing & Fabrication:

MIG Welding, TIG Welding, Stick Welding, 3D Printing, Surface Grinding, Band Saw Operation, Shop Safety Procedures

Engineering:

Mechanical Design, Engineering Drawings, Technical Documentation, Additive Manufacturing, Design Analysis

Professional:

Project Coordination, Leadership, Team Collaboration, Problem Solving, Bilingual Communication (English & Spanish)